In-Person

Genomics Compute Cluster

STAR RNA-seq Aligner

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Setup...

1. log onto Quest

   ssh <netid>@login.quest.northwestern.edu

   # enter your netid password (it won't look like you're typing)

2. move to our classroom folder

   cd /projects/e32680

3. make your own subfolder if you don’t have one

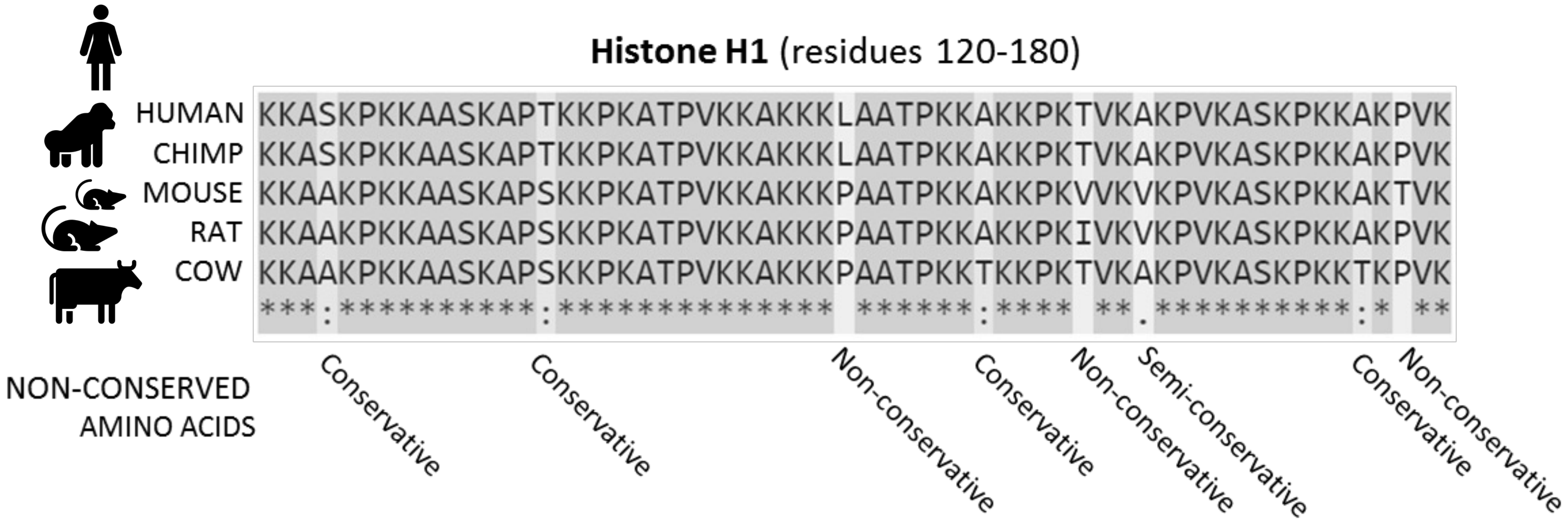
   mkdir <folder_name>

4. move into your folder

   cd <folder_name>

Class materials are at:
github.com/nuitrcs/rna_alignment

What is alignment? - hypothesis of homology



How to pick an aligner?

- Type of data you have
  - aligning short read data (Illumina)
  - aligning long read data (PacBio, Nanopore)
  - aligning RNA-seq data - splice aware
- Accuracy
  - percent of reads aligned
  - estimated gene coverage
- Runtime

Comparison of Short-Read Sequence Aligners Indicates Strengths and Weaknesses for Biologists to Consider

*Ryan Musich[1], Lance Cadle-Davidson[1,2] and Michael V. Osier[1]**

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[2]USDA-Agricultural Research Service, Grape Genetics Research Unit, Geneva, NY, United States

RESEARCH ARTICLE **Open Access**

Benchmarking short sequence mapping tools

Ayat Hatem[1,2], Doruk Bozdağ[2], Amanda E Toland[3] and Ümit V Çatalyürek[1,2*]

A Recent (2020) Comparative Analysis of Genome Aligners Shows HISAT2 and BWA are Among the Best Tools

Article

Aligning the Aligners: Comparison of RNA Sequencing Data Alignment and Gene Expression Quantification Tools for Clinical Breast Cancer Research

Isaac D. Raplee [1], Alexei V. Evsikov [2] and Caralina Marín de Evsikova [1,2,*]

Why does it matter if you have DNA or RNA data?

often aligning to a reference genome

- create a standard index to map to

most RNA-seq data is mature mRNA

- typically no introns in the sequence of reads

but genomes have introns!

- might introduce long gaps in the alignment, which are generally penalized against when determining the `best` mapping of a read
- might show two or more partial matches for a read to different exons

spice-aware aligners

- allows identification of novel splice junctions, regions of expression, structural variants...

STAR - Spliced Transcripts Alignment to a Reference

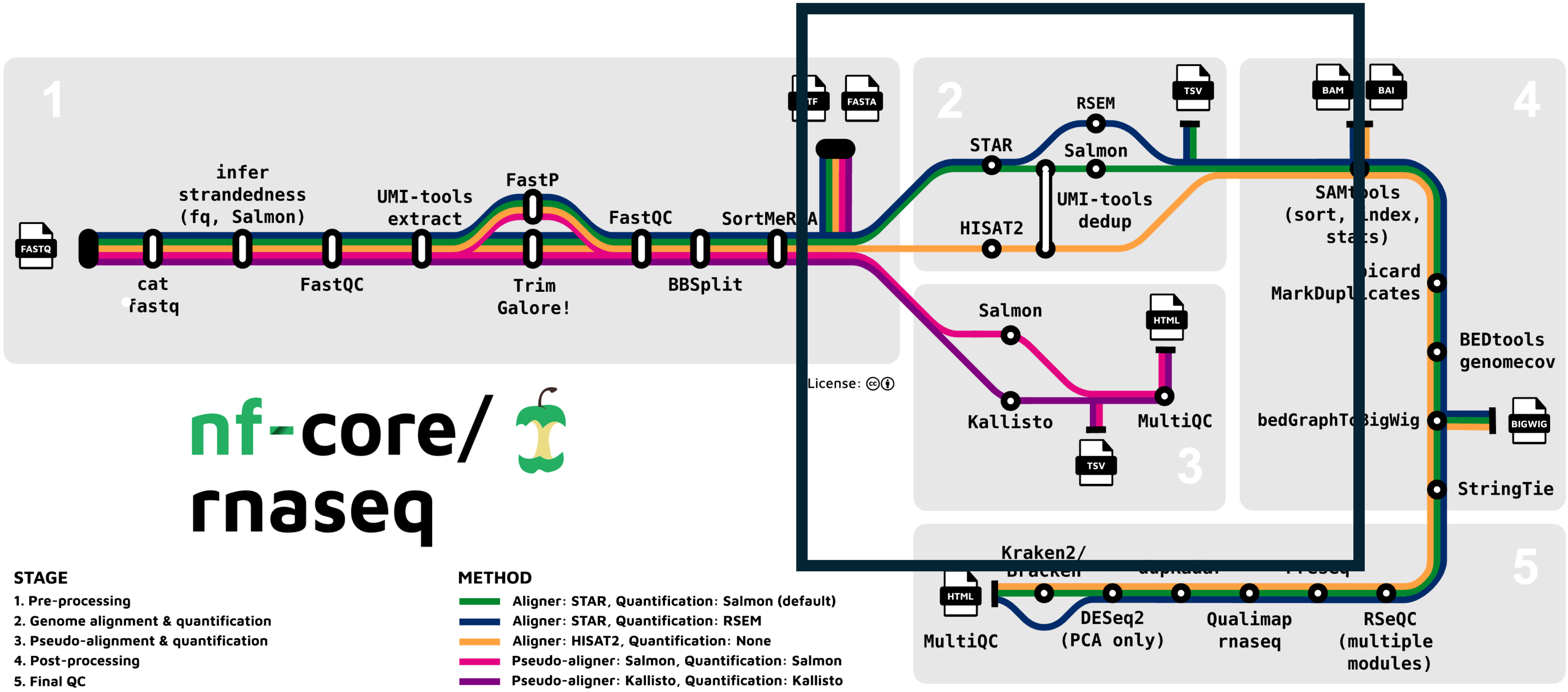
- **splice aware** aligner
- seed-and-extend (similar to bwa) based on compressed suffix arrays
- potentially better suited for draft or low-quality genomes than HISAT2 or other splice-aware options
- widely used, and default option of many pipelines, include nf-core's RNA-seq

- Resources:
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3530905/
- https://github.com/alexdobin/STAR

HISAT2 - Hierarchical Indexing for Spliced Alignment of Transcripts

- **splice aware** aligner
- graph-based approach (Ferragina Manzini index)
- less resource intensive than STAR
- widely used, and available in nf-core's RNA-seq pipeline

- Resources:
- https://daehwankimlab.github.io/hisat2/
- https://daehwankimlab.github.io/hisat2/manual/



1
FASTQ
cat fastq
infer strandedness (fq, Salmon)
FastQC
UMI-tools extract
FastP
Trim Galore!
FastQC
BBSplit
FASTA
2
STAR
RSEM
Salmon
TSV
HISAT2
UMI-tools dedup
Salmon
HTML
Kallisto
MultiQC
TSV
3
License:
BAM
BAI
4
BEDtools genomecov
BIGWIG
StringTie
Kraken2/
HTML
MultiQC
DESeq2 (PCA only)
Qualimap rnaseq
RSeQC (multiple modules)
5
nf-core/ rnaseq
STAGE
1. Pre-processing
2. Genome alignment & quantification
3. Pseudo-alignment & quantification
4. Post-processing
5. Final QC
METHOD
Aligner: STAR, Quantification: Salmon (default)
Aligner: STAR, Quantification: RSEM
Aligner: HISAT2, Quantification: None
Pseudo-aligner: Salmon, Quantification: Salmon
Pseudo-aligner: Kallisto, Quantification: Kallisto

How do we use it?

- Generate genome index files
- Map reads to the genome

Generate genome index files

- if you are generating your own index you need:
  1. fasta file of the genome sequence
  2. gtf files of the genome annotation

     - an annotation file is necessary for splice-aware aligners because part of the indexing process is building a database of splice junctions

Generate genome index files

```
STAR \
  --runMode genomeGenerate \
  --runThreadN 4 \
  --genomeDir /path/to/genomeDir \
  --genomeFastaFiles /path/to/genome.fa \
  --sjdbGTFfile /path/to/genome.gtf \
  --sjdbOverhang <ReadLength-1>
```

Generate genome index files

```
STAR \
  --runMode genomeGenerate \
  --runThreadN 4 \  # number of cores to parallelize over
  --genomeDir /path/to/genomeDir \  # output location
  --genomeFastaFiles /path/to/genome.fa \  # input location
  --sjdbGTFfile /path/to/genome.gtf \  # input location
  --sjdbOverhang <ReadLength-1> # length of region around
                                  annotation to be used for
                                  constructing splice junctions
                                  database, default is 100,
                                  ReadLength-1 is recommendation
```

Running this on Quest...

- Quest uses a scheduler to control access to compute resources.
- We will wrap the STAR command in a ‘job script’ which will request compute resource through the scheduler and set up the software environment for STAR to run.
- I have a template job script available to you! Please copy it into your folder with the following:

```
cd /projects/e32680 # you may already be here
cp 05_rnaseq_alignment/star1.sh <your_folder>
cd <your_folder>
ls # you should see star1.sh
```

Running this on Quest…

Open the script with nano:

```
nano star1.sh
```

Change the genomeDir to:

```
/projects/e32680/<your_folder>/STARindex
```

Save and exit nano with CTRL+O, ENTER, CTRL+X

Launch the job with:

```
sbatch star1.sh
```

Generate genome index files

We host pre-generated index files on Quest for many model systems:

- /projects/genomicsshare/AWS_iGenomes/references/Mus_musculus/Ensembl/GRCm38/Sequence/STARIndex
- /projects/genomicsshare/AWS_iGenomes/references/Homo_sapiens/Ensembl/GRCh37/Sequence/STARIndex

*These were generated with a slightly older version of STAR, so you will want to use the STAR/2.5.2 module. Newer versions coming soon!

Map reads to genome

```
STAR \
  --runThreadN 8 \  # number of cores to parallelize over
  --genomeDir /path/to/genomeDir \  # input location
  --readFilesIn /path/to/genome.fa  # input location, this can
                                    be multiple files fasta or
                                    fastq, both paired end read
                                    files need to be supplied
```

Those are the basic options but there are about a million more you can adjust...

Copy an alignment script to your folder….

```
cp /projects/e32680/05_rnaseq_alignment/star2.sh .
```

Copy an alignment script to your folder….

```
cp /projects/e32680/05_rnaseq_alignment/star2.sh .
```

What does this script contain?

star2.sh

```
STAR \
  --runThreadN 4 \
  --genomeDir /projects/e32680/05_rnaseq_alignment/STARindex
\
  --readFilesIn
/projects/e32680/05_rnaseq_alignment/SRR7773980_1.fastq.gz,/p
rojects/e32680/02_rnaseq_alignment/SRR7773980_2.fastq.gz \
  --readFilesCommand gunzip -c \
  --outFileNamePrefix ./STARoutputs
```

Map reads to genome:

Edit the genomeDir to the location of your index:

```
/projects/e32680/<your_folder>/STARindex
```

Save and exit nano with CTRL+O, ENTER, CTRL+X

Launch the job with:

```
sbatch star2.sh
```

*From <u>your</u> folder! Not the 05_rnaseq_alignment one.

Outputs

```
(base) [hsc945@quser44 STARoutputs]$ ls
Aligned.out.sam  Log.final.out  Log.out  Log.progress.out  SJ.out.tab
```

- Aligned.out.sam: Alignment file in SAM format
- Log.final.out: Log output that includes higher level summaries of mapping rates, number of splice sites, and runtime information
- Log.out: Log output that includes step by step detailed output
- Log.progress.out: Log output that checks progress and reports summary every minute during runtime
- SJ.out.tab: Splice junction output in tab delimited format

Aligned.out.sam

- alignment file in SAM (sequence alignment map) format
- output format can be controlled with

```
--outSAMtype
```

- for unsorted BAM (binary alignment map) format:

```
--outSAMtype BAM Unsorted
```

- for sorted BAM format:

```
--outSAMtype BAM SortedByCoordinate
```

- for both files:

```
--outSAMtype BAM Unsorted SortedByCoordinate
```

`--outBAMsortingThreadN` # controls number of threads for sorting, default is 6

SJ.out.tab

- high confidence splice junctions in tab-delimited format
- 9 columns: chromosome, first base of intron, last base of intron, strand, intron motif, annotation, number of uniquely mapped reads crossing junction, number of multi-mapping reads crossing junction, maximum spliced alignment overhang
- filtering can be controlled with

```
--outSJfilter*
```

- for only uniquely mapping reads:

```
--outSJfilterReads Unique
```

Exercise options:

1. Translate the methods blurb on the following slide to a job script for Quest.
2. Convert star2.sh to run hisat2.sh instead. The relevant flags are:

   -p or --threads for processors, -x for genome index, -1 for read1, -2 for read2, -S for name and location of sam file output
3. Map the two fastq files in /projects/e32680/05_rnaseq_alignment/data to the Star Index for GRCm39 located at /projects/e32680/05_rnaseq_alignment/GCA_000001635.9_STARIndex/STARIndex

”Pioneer factor GATA6 promotes colorectal cancer through 3D genome regulation”

Lyu et al. 2025. *Science Advances*

https://pmc.ncbi.nlm.nih.gov/articles/PMC11804904/#sec11

Materials and Methods

RNA-seq data processing

> The trimmed FASTQ files of colon cancer cell lines are aligned against hg38 human reference genome using STAR (65)(v.2.7.10b) with parameters “--outSAMunmapped Within --outFilterType BySJout --outSAMattributes NH HI AS NM MD --outFilterMultimapNmax 20 --outFilterMismatchNmax 999 --outFilterMismatchNoverReadLmax 0.04 --alignIntronMin 20 --alignIntronMax 1000000 --alignMatesGapMax 1000000 --alignSJoverhangMin 8 --alignSJDBoverhangMin 1 --sjdbScore 1”. RSEM (66)(v1.3.3) facilitated the quantification and expression calculation for known genes with GENCODE v33. Genes with TPM value < 1 in all samples were excluded from downstream analyses.